

**S.E. (Computer Engineering)**  
**Discrete Structure (2012 Course)**  
**(Semester - I)**

**Time: 2 Hours****Max. Marks : 50****Instructions to the candidates:**

- 1) Attempt Q1 or Q2 , Q3 or Q4, Q5 or Q6 , Q7 or Q8
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of Calculator is allowed.
- 5) Assume Suitable data if necessary

- Q1) a) With the help of mathematical induction prove that, [4]

$$1^2 + 3^2 + 5^2 + (2n - 1)^2 = \frac{n(2n-1)(2n+1)}{3}$$

- b) Over the universe of book defined propositions [2]

B(x): x has blue cover

M(x): x is maths book

I(x): x published in India

Translate the following

- i)  $\forall x(M(x) \wedge I(x) \rightarrow B(x))$
- ii) There are maths books published outside India.

- c) Let  $x = \{1, 2, \dots, 7\}$  and  $R = \{(x, y) | x - y \text{ is divisible by } 3\}$ . Show that R is equivalence relation. Draw graph of R. [6]

- Q2) a) Prove the following using venn diagram [2]

$$A \cap (B \oplus C) = (A \cap B) \oplus (A \cap C)$$

- b) Among the integers 1 to 1000 [4]
- i) How many of them are not divisible by 3 nor by 5 nor by 7.
  - ii) How many are not divisible by 5 or 7 but divisible by 3.

- c) Let  $x = \{2, 3, 6, 12, 24, 36\}$   $x \leq y$  if x divides y [6]

Find

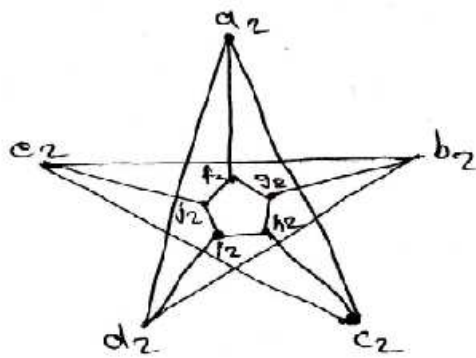
1. Maximal element
2. Minimal element
3. Chain
4. Antichain
5. Is poset lattice ?

- Q3) a) Define the following [6]

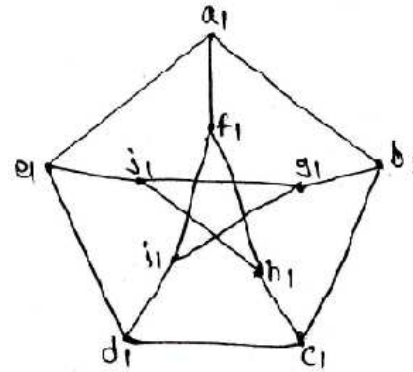
1. Group
2. Ring
3. Field

b) Show that the following graphs are isomorphic

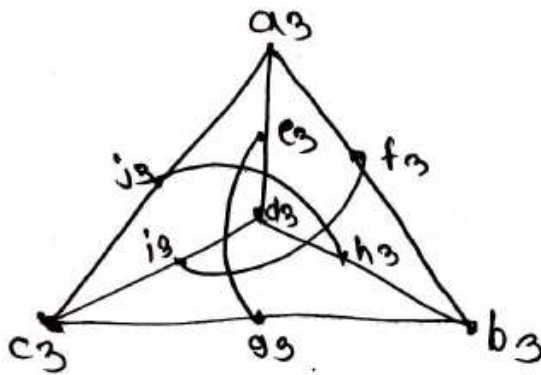
[6]



G1



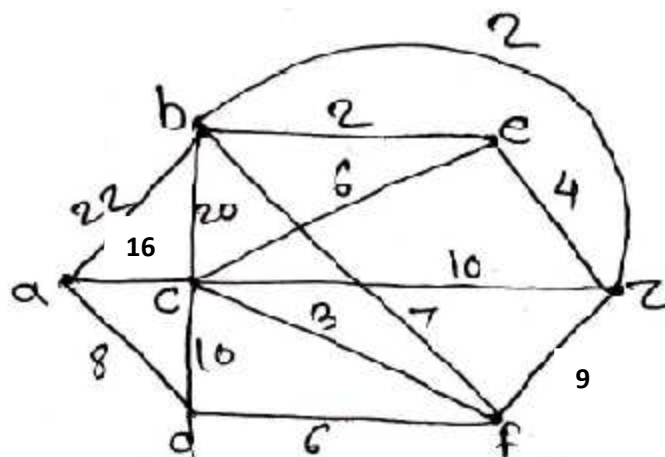
G2



G3

Q4) a) Find the shortest distance in the given figure from a to z by using Dijkstra shortest path algorithm

[6]



b) Prove that the set  $Z$  of all integers with binary operation  $*$  defined by  $a * b = a + b + 1$  such that  $\forall a, b \in Z$  is an abelian group

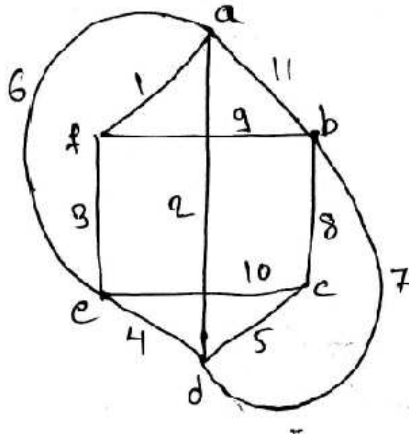
[6]

Q5) a) For the following set of weights, construct optimal binary prefix code. For each weight in the set, give the corresponding code words. 10, 30, 05, 15, 20, 15, 05.

[7]

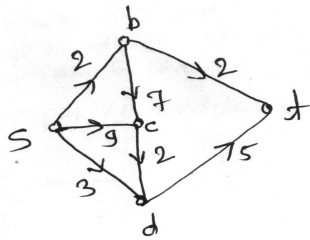
b) Find the minimum spanning tree of the given figure using Kruskal's algorithm

[6]



- Q6) a) Define the following terms with reference to tree with example [6]
- Level and height of a tree
  - M-ary tree
  - Eccentricity of a vertex

- b) Find the maximum flow of the transport network given in the figure. [7]



- Q7) a) A husband and wife appear in an interview for two vacancies in the same post. The probability of husband's selection is  $\frac{1}{7}$  and that of wife's selection is  $\frac{1}{5}$ . what is the probability that [7]
- both of them will be selected
  - only one of them will be selected
  - none of them will be selected

- b) How many numbers of 7 digits can be formed with the digits 0,2,2,5,6,6,6 . How many of them are even ? [6]

- Q8) a) A committee of 5 people is to be formed from a group of 4 men and 7 women. How many possible committees can be formed if at least 3 women are on the committee? [6]

- b) A box contains 6 red, 4 white and 5 black balls. A person draws 4 balls from the box at random. Find the probability that among the balls drawn there is at least one ball of each color. [7]